Module 2: Data Analytics Fundamentals

📖 Overview

Before building dashboards and reports in Power BI, it is essential to understand the fundamentals of data analytics. This module introduces the core concepts of data, analytics, databases, data warehouses, and analytical thinking.

Data analytics is the foundation of Business Intelligence because reports and dashboards are only as valuable as the insights they provide.

🎯 Learning Objectives

After completing this module, you will be able to:

- Understand the fundamentals of Data Analytics
- Differentiate between data and information
- Understand different types of analytics
- Learn common data types
- Understand structured and unstructured data
- Understand databases and data warehouses
- Learn the analytics lifecycle
- Understand the role of data analysts in organizations

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1️⃣ What is Data Analytics?

Definition

Data Analytics is the process of examining, cleaning, transforming, and interpreting data to discover meaningful patterns, trends, and insights that support decision-making.

Simple Example

Imagine an online store has the following sales data:

Month	Sales
January	₹50,000
February	₹60,000
March	₹75,000

Raw numbers alone are data.

Analyzing these numbers to determine sales growth is analytics.

Concluding that sales are increasing because of a marketing campaign is an insight.

Main Goal of Data Analytics

Transform:

```
Data
 ↓
Information
 ↓
Insights
 ↓
Business Decisions
```

2️⃣ Why Data Analytics Matters

Organizations generate large amounts of data every day.

Examples:

- Customer transactions
- Website visits
- Employee records

- Inventory data
- Marketing campaign results

Without analytics:

- Data remains unused
- Decision-making becomes difficult

With analytics:

- Trends become visible
- Opportunities are identified
- Risks are reduced

Benefits of Data Analytics

Better Decision Making

Organizations make data-driven decisions instead of assumptions.

Cost Reduction

Identifies inefficiencies and waste.

Revenue Growth

Discovers profitable opportunities.

Customer Understanding

Analyzes customer behavior and preferences.

Competitive Advantage

Provides insights faster than competitors.

3️⃣ Data vs Information vs Insight

Understanding these concepts is critical.

Data

Raw facts without context.

Example:

```
100
150
```

```
200
```

These numbers alone have little meaning.

Information

Data organized into a meaningful format.

Example:

```
January Sales = 100
February Sales = 150
March Sales = 200
```

Now the data has context.

Insight

An actionable conclusion derived from information.

Example:

```
Sales increased by 100% in three months.
Marketing efforts may be contributing to growth.
```

4️⃣ Types of Analytics

There are four major types of analytics.

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1. Descriptive Analytics

Question Answered

```
What happened?
```

Example

Monthly Sales Report

Month	Sales
January	₹50,000
February	₹60,000

Descriptive analytics summarizes historical data.

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2. Diagnostic Analytics

Question Answered

```
Why did it happen?
```

Example

Sales increased because:

- Marketing campaign succeeded
- Website traffic increased

Diagnostic analytics identifies causes.

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3. Predictive Analytics

Question Answered

```
What is likely to happen?
```

Example

Forecast next month's sales using historical trends.

Predictive analytics uses statistical models and machine learning.

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4. Prescriptive Analytics

Question Answered

```
What should we do?
```

Example

Recommend increasing advertising budget.

Prescriptive analytics suggests actions.

Analytics Maturity Model

```
Descriptive
     ↓
Diagnostic
     ↓
Predictive
     ↓
Prescriptive
```

5️⃣ Data Types

Understanding data types is important for reporting and analysis.

Numeric Data

Contains numbers.

Examples:

```
100
5000
25.75
```

Used for calculations.

Text Data

Contains characters and words.

Examples:

```
Delhi
Laptop
Customer Name
```

Date & Time Data

Examples:

```
01-Jan-2026
12:30 PM
```

Used for trend analysis.

Boolean Data

Contains only:

```
TRUE
FALSE
```

Examples:

- Is Active?
- Is Paid?

Currency Data

Examples:

```
₹1000
$500
€300
```

Used in financial reporting.

6️⃣ Structured vs Unstructured Data

Structured Data

Organized into rows and columns.

Example:

Customer ID	Name
101	Rahul
102	Priya

Sources:

- SQL Databases
- Excel
- CSV Files

Unstructured Data

Does not follow a predefined format.

Examples:

- Images
- Videos
- Audio Files
- Social Media Posts
- Emails

Semi-Structured Data

Partially organized.

Examples:

- JSON
- XML

Comparison

Structured	Unstructured
Organized	Not Organized
Easy to Analyze	More Complex
SQL Databases	Images, Videos
Excel Files	Social Media Content

7️⃣ Databases

What is a Database?

A database is an organized collection of data stored electronically.

Example

Customer Database

CustomerID	Name	City
1	Rahul	Delhi
2	Priya	Mumbai

Popular Databases

- SQL Server
- MySQL
- PostgreSQL
- Oracle
- SQLite

Advantages

- Fast retrieval
- Data security
- Data consistency
- Multi-user access

8️⃣ Data Warehouses

What is a Data Warehouse?

A Data Warehouse is a centralized repository that stores data from multiple sources for reporting and analysis.

Purpose

Operational databases are optimized for transactions.

Data warehouses are optimized for analytics.

Characteristics

Subject-Oriented

Organized around business subjects.

Example:

- Sales
- Finance
- HR

Integrated

Combines multiple data sources.

Historical

Stores long-term data.

Non-Volatile

Data remains stable after loading.

9️⃣ Data Lakes

What is a Data Lake?

A Data Lake stores large amounts of raw data in its original format.

Data Warehouse vs Data Lake

Data Warehouse	Data Lake
Processed Data	Raw Data
Structured	Structured & Unstructured
Reporting	Advanced Analytics
Faster Queries	Flexible Storage

🔟 Analytics Lifecycle

Data analytics follows a structured process.

Step 1: Business Understanding

Identify business objectives.

Example:

```
Why are sales decreasing?
```

Step 2: Data Collection

Gather data from:

- Databases
- APIs
- Excel Files
- Applications

Step 3: Data Cleaning

Remove:

- Missing Values
- Duplicates
- Errors

Step 4: Data Analysis

Identify:

- Trends
- Patterns
- Relationships

Step 5: Visualization

Create:

- Reports
- Dashboards
- Charts

Step 6: Decision Making

Use insights to take action.

1️⃣1️⃣ Data Analyst Roles & Responsibilities

A Data Analyst typically performs:

Data Collection

Gathering data from various sources.

Data Cleaning

Preparing data for analysis.

Data Analysis

Finding trends and insights.

Data Visualization

Creating dashboards and reports.

Business Communication

Presenting findings to stakeholders.

1️⃣2️⃣ Common Analytics Tools

Spreadsheet Tools

- Microsoft Excel
- Google Sheets

Database Tools

- SQL Server
- MySQL
- PostgreSQL

Visualization Tools

- Power BI
- Tableau
- Looker Studio

Programming Languages

- Python
- R

Cloud Platforms

- Microsoft Azure
- AWS
- Google Cloud Platform

📌 Key Terminologies

Term	Definition
Data	Raw facts
Information	Organized data
Insight	Actionable understanding
Analytics	Process of extracting insights
Database	Organized data storage
Data Warehouse	Analytics-focused repository
Data Lake	Raw data storage
KPI	Key Performance Indicator
Dashboard	Visual summary of metrics
Visualization	Graphical representation of data

📝 Practice Exercises

Exercise 1

Identify examples of:

- Structured Data
- Unstructured Data
- Semi-Structured Data

Exercise 2

Choose a company and list:

- Data Sources

- Possible KPIs
- Business Decisions

Exercise 3

Explain the difference between:

- Database
- Data Warehouse
- Data Lake

Exercise 4

Research the four types of analytics and provide one real-world example of each.

Exercise 5

Create a diagram showing the Analytics Lifecycle.

🎯 Mini Project

Retail Sales Analysis Planning

Imagine you are working for a retail company.

Task

Identify:

- Business Problem
- Data Sources
- KPIs
- Expected Insights

Deliverable

Prepare a one-page analytics plan.

📚 Module Summary

In this module, you learned:

✅ Fundamentals of Data Analytics

✅ Data vs Information vs Insight

✅ Four Types of Analytics

✅ Data Types

✅ Structured, Unstructured & Semi-Structured Data

✅ Databases

✅ Data Warehouses

✅ Data Lakes

✅ Analytics Lifecycle

✅ Data Analyst Responsibilities

These concepts form the analytical foundation required before working with data sources, transformations, and modeling in Power BI.

⏭️ Next Module

Module 3: Data Sources & Data Connectivity in Power BI

Topics Covered:

- Understanding Data Sources
- Import Mode
- DirectQuery
- Live Connection
- Connecting Excel, CSV, SQL Server & Web Data
- Data Refresh Concepts